Instrument list + I/O count

SO3 Converter Ver 1 · SO3-V1-IN-INL-001 · Rev 0

Department of Chemical Engineering, Chulalongkorn University (teaching)

REV	DATE	DESCRIPTION	ORIGINATOR	CHECKED / APPROVED
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Generated for teaching from the SO3 Converter Ver 1 operator training simulator. Every number comes from the simulator model (design point, noise off) or is marked assumed. Not for construction. No client, contractor or vendor is represented.

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1 · Purpose and scope

This document lists every instrument of the SO3 Converter Ver 1 converter section (battery limits: dried air in, molten sulfur in, acid and steam out, tail gas to stack) with its service, type, range, normal value, alarm and trip settings, control system, I/O type and fail position, and counts the I/O per system. It also records the DCS / SIS architecture assumed for the unit and a short control narrative.

Sources: the simulator model (design point, noise off: snapDesign, makeCtrls(), tripConditions(), bmsStep(), K), the DCS tag tables (src/ui1.js IND, BUB, ALM, CRANGE) and the EPC P&ID sheets SO3-PID-001/002 rev A. Ranges are DCS display spans unless stated (no instrument datasheet exists: EQ-ei-02). Tags or I/O marked assumed are typical practice, not in the model.

Related documents (not repeated here): SO3-PID-001/002 rev A (P&ID, loop and interlock schedules); SO3-V1-IN-CE-001 (cause & effect); SO3-V1-IN-SRS-001 (SRS); SO3-V1-SA-HZP-001 (HAZOP report); control loop design, safety layers and LOPA studies on the simulator tabs.

2 · Instrument list

88 tags. Normal = model at design. Alarms from the DCS alarm table; trips from the SIS constants with the logic-solver delays. Fail: FO fail open, FC fail closed, LC locked closed. P&ID: sheet of SO3-PID rev A, — = not drawn.

Tag	Loop	Service	Type	Range	Unit	Normal	Alm L	Alm H	Trip	Sys	I/O	Fail	P&ID	Note
FE-100	FI-100	Total air from K-101, orifice	FE	—		—	—	—	—	Local	—	—	001	Orifice drawn on the EPC sheet
FT-100	FI-100	Total air from K-101	FT	0–300	kNm³/h	239	200	265	—	DCS	AI	—	001	Also the sensor of FSLL-100 and of the H-101 low-air interlock
FSLL-100	FSLL-100	Low total air: sulfur shut-off (I-1)	SIF	0–300	kNm³/h	239	—	—	< 90.0 (3 s)	SIS	AI	—	001	SIS input taken from FT-100 (shared with the DCS indication)
FSL-109	FSL-109	H-101 low-air interlock (burner package, < 60 kNm³/h)	SOFT	0–300	kNm³/h	239	—	—	< 60.0	Local	—	—	—	Tag assumed; in the model a local burner interlock, not an SIS function
PT-101	PI-101	K-101 blower discharge pressure	PT	0–80	kPa g	60.0	—	—	—	DCS	AI	—	001	
FE-101	FIC-101	Combustion air to F-101, orifice	FE	—		—	—	—	—	Local	—	—	001	
FT-101	FIC-101	Combustion air to F-101	FT	0–300	kNm³/h	204	—	—	—	DCS	AI	—	001	PV of FIC-101
FFC-101	FFC-101	Air/sulfur ratio station (remote SP of FIC-101)	SOFT	3–8	kNm³/t	4.99	—	—	—	DCS	—	—	001	DCS block, no field I/O
FV-101	FIC-101	Combustion-air control valve	CV	0–100	% open	54.9	—	—	—	DCS	AO	FO	001	
FE-102	FIC-102	Dilution air to converter inlet, orifice	FE	—		—	—	—	—	Local	—	—	001	
FT-102	FIC-102	Dilution air to converter inlet	FT	0–80	kNm³/h	34.1	—	—	—	DCS	AI	—	001	PV of FIC-102
FV-102	FIC-102	Dilution-air control valve	CV	0–100	% open	45.0	—	—	—	DCS	AO	FO	001	
AT-104	AIC-104	SO_2 in the gas to bed 1	AT	0–16	% SO_2	12.0	11.0	12.8	—	DCS	AI	—	001	PV of AIC-104 (cascade master of FIC-102)
AT-105	AI-105	O_2 in the gas to bed 1	AT	0–21	% O_2	9.00	8.00	—	—	DCS	AI	—	001	
PT-200	PI-200	Converter inlet pressure	PT	0–60	kPa g	35.0	—	42.0	—	DCS	AI	—	—	On the DCS, on neither drawing (P&ID tab part 5)
FE-103	FIC-103	Molten sulfur to F-101, flow element	FE	—		—	—	—	—	Local	—	—	001	Steam-jacketed
FT-103	FIC-103	Molten sulfur to F-101	FT	0–60	t/h	40.9	35.0	44.0	—	DCS	AI	—	001	PV of FIC-103 and input of FFC-101
FV-103	FIC-103	Sulfur flow control valve	CV	0–100	% open	50.0	—	—	—	DCS	AO	FC	001	Forced to MAN 0 % by every SIS trip (through the DCS)
XV-103	XV-103	Sulfur shut-off valve (SIS final element)	XV	—		—	—	—	—	SIS	—	FC	001	Spring-return, de-energise to trip (SRS common requirements)
XY-103	XV-103	Solenoid of XV-103	RLY	—		—	—	—	—	SIS	DO	FC	001	Tag assumed (solenoid drawn on XV-103)
ZSC-103	XV-103	XV-103 closed limit switch	ZS	—		—	—	—	—	SIS	DI	—	—	Assumed typical: valve position proven to the SIS
ZSO-103	XV-103	XV-103 open limit switch	ZS	—		—	—	—	—	SIS	DI	—	—	Assumed typical

Tag	Loop	Service	Type	Range	Unit	Normal	Alm L	Alm H	Trip	Sys	I/O	Fail	P&ID	Note
TE/TT-106	TI-106	Sulfur furnace F-101 outlet	TE/TT	0–1300	°C	1121	—	—	—	DCS	AI	—	001	Start permissive ≥ 450 °C
TE/TT-107	TI-107	Waste heat boiler B-101 gas outlet	TE/TT	0–700	°C	274	—	340	—	DCS	AI	—	001	
FT-108	FI-108	Saturated steam from B-101	FT	0–150	t/h	94.0	—	—	—	DCS	AI	—	001	
FT-109	FI-109	H-101 preheat burner firing (fuel meter, scaled in MW)	FT	0–30	MW	0.00	—	—	—	DCS	AI	—	—	On the reference drawing, not on the EPC sheets; fuel meter type assumed
HV-109	HIC-109	H-101 firing valve (burner package)	CV	0–100	% open	0.00	—	—	—	DCS	AO	FC	001	Tag assumed (model: H-101 firing output)
TV-201	TIC-201	B-101 hot-gas bypass valve	CV	0–100	% open	34.6	—	—	—	DCS	AO	FC	001	
TE/TT-201	TIC-201	Bed 1 inlet	TE/TT	300–550	°C	420	405	435	—	DCS	AI	—	001	PV of TIC-201; start permissive ≥ 400 and ≤ 440 °C
TE/TT-202	TI-202	Bed 1 middle (catalyst)	TE/TT	300–700	°C	632	—	—	—	DCS	AI	—	001	Start permissive ≥ 400 °C
TE/TT-203	TI-203	Bed 1 exit	TE/TT	300–700	°C	631	—	640	—	DCS	AI	—	001	One element TE-203 serves TI-203, its alarm and TSHH-203
TSHH-203	TSHH-203	Bed 1 exit high-high: sulfur shut-off (I-2)	SIF	300–700	°C	631	—	—	> 660 (5 s)	SIS	AI	—	001	SIS input from TE-203 (shared with the DCS indication and alarm)
PDT-204	PDI-204	Bed 1 pressure drop	PDT	0–20	kPa	4.52	—	8.00	—	DCS	AI	—	001	
TE/TT-207	TI-207	Superheated steam leaving SH-201	TE/TT	200–600	°C	292	—	—	—	DCS	AI	—	—	On the DCS, on neither drawing
TV-231	TIC-231	SH-201 gas bypass valve	CV	0–100	% open	38.9	—	—	—	DCS	AO	FO	001	TIC-231 drawn at TV-231 on sheet 1
TE/TT-211	TIC-211	Bed 2 inlet	TE/TT	300–600	°C	445	430	460	—	DCS	AI	—	001	PV of TIC-211
TV-211	TIC-211	E-201 hot-side bypass valve	CV	0–100	% open	32.0	—	—	—	DCS	AO	FO	001	
TE/TT-212	TI-212	Bed 2 middle	TE/TT	300–700	°C	528	—	—	—	DCS	AI	—	001	
TE/TT-213	TI-213	Bed 2 exit	TE/TT	300–700	°C	528	—	—	—	DCS	AI	—	001	
PDT-214	PDI-214	Bed 2 pressure drop	PDT	0–10	kPa	2.86	—	—	—	DCS	AI	—	001	
TE/TT-253	TI-253	E-201 hot-side outlet (before the TV-211 bypass joins)	TE/TT	200–700	°C	406	—	—	—	DCS	AI	—	—	On the DCS, on neither drawing
TE/TT-221	TIC-221	Bed 3 inlet	TE/TT	300–600	°C	440	425	455	—	DCS	AI	—	001	PV of TIC-221
TV-221	TIC-221	E-202 hot-side bypass valve	CV	0–100	% open	35.2	—	—	—	DCS	AO	FO	001	
TE/TT-222	TI-222	Bed 3 middle	TE/TT	300–700	°C	464	—	—	—	DCS	AI	—	001	
TE/TT-223	TI-223	Bed 3 exit	TE/TT	300–700	°C	465	—	—	—	DCS	AI	—	001	
PDT-224	PDI-224	Bed 3 pressure drop	PDT	0–10	kPa	3.16	—	—	—	DCS	AI	—	001	
TE/TT-254	TI-254	Return gas leaving E-202 (to E-201)	TE/TT	0–600	°C	204	—	—	—	DCS	AI	—	—	On the DCS, on neither drawing
TE/TT-231	TIC-231	Bed 4 inlet (return gas)	TE/TT	300–550	°C	425	405	445	—	DCS	AI	—	002	TT-231 drawn on sheet 2 apart from TIC-231 (sheet note 4)
TE/TT-232	TI-232	Bed 4 middle	TE/TT	300–700	°C	438	—	—	—	DCS	AI	—	002	Start permissive ≥ 360 °C
TE/TT-233	TI-233	Bed 4 exit	TE/TT	300–700	°C	441	—	—	—	DCS	AI	—	002	
PDT-234	PDI-234	Bed 4 pressure drop	PDT	0–10	kPa	2.54	—	—	—	DCS	AI	—	002	
TE/TT-241	TIC-241	Bed 5 inlet	TE/TT	300–550	°C	428	410	445	—	DCS	AI	—	002	PV of TIC-241
TV-241	TIC-241	E-204 gas bypass valve	CV	0–100	% open	43.7	—	—	—	DCS	AO	FO	002	
TE/TT-242	TI-242	Bed 5 middle	TE/TT	300–700	°C	428	—	—	—	DCS	AI	—	002	
TE/TT-243	TI-243	Bed 5 exit	TE/TT	300–700	°C	427	—	—	—	DCS	AI	—	002	
PDT-244	PDI-244	Bed 5 pressure drop	PDT	0–10	kPa	2.50	—	—	—	DCS	AI	—	002	
HV-112	HV-112	Start-up hot-gas line to bed 4 (locked closed)	HV	—		—	—	—	—	Local	—	LC	001	LC hand valve
ZSC-112	HV-112	HV-112 closed limit switch (start permissive)	ZS	—		—	—	—	—	DCS	DI	—	—	Assumed: the permissive "HV-112 closed" needs a proven position
TE/TT-251	TIC-251	Gas to IPA T-301	TE/TT	100–350	°C	190	165	230	—	DCS	AI	—	002	PV of TIC-251

Tag	Loop	Service	Type	Range	Unit	Normal	Alm L	Alm H	Trip	Sys	I/O	Fail	P&ID	Note
TV-251	TIC-251	E-203 gas bypass valve	CV	0–100	% open	30.0	—	—	—	DCS	AO	FO	002	
FT-253	FI-253	BFW to economizers E-203/E-204/E-205	FT	0–200	t/h	150	110	—	—	DCS	AI	—	—	On the reference drawing, not on the EPC sheets
TE/TT-255	TI-255	BFW leaving E-203	TE/TT	0–300	°C	243	—	—	—	DCS	AI	—	—	On the DCS, on neither drawing
TE/TT-305	TI-305	Return gas leaving T-301 (to E-202)	TE/TT	0–300	°C	83.0	—	—	—	DCS	AI	—	002	
FT-301	FI-301	IPA acid circulation	FT	0–2000	m³/h	1600	1200	—	—	DCS	AI	—	002	Also the sensor of FSLL-301 and of the P-301B auto-start; meter type assumed (magnetic)
FSLL-301	FSLL-301	IPA acid circulation low-low: sulfur shut-off (I-3)	SIF	0–2000	m³/h	1600	—	—	< 800 (10 s)	SIS	AI	—	002	SIS input from FT-301 (shared with the DCS indication, alarm and auto-start)
FSL-301	FSL-301	P-301B auto-start on low IPA acid (< 60 % for 3 s)	SOFT	0–2000	m³/h	1600	—	—	< 960 (3 s)	DCS	—	—	002	DCS logic; disarms after one attempt
HS-301B	HS-301B	P-301B auto-start arm / disarm	SOFT	—		—	—	—	—	DCS	—	—	002	Soft switch on the DCS
AT-303	AI-303	IPA acid strength	AT	95–100	% H_2SO_4	98.5	98.0	—	—	DCS	AI	—	002	
TE/TT-304	TI-304	IPA acid temperature to T-301	TE/TT	0–150	°C	80.0	—	95.0	—	DCS	AI	—	002	
TE/TT-252	TI-252	Gas to the final absorber T-302	TE/TT	0–500	°C	160	—	—	—	DCS	AI	—	002	
FT-302	FI-302	FAT acid circulation	FT	0–1500	m³/h	1000	750	—	—	DCS	AI	—	002	Also the sensor of FSLL-302
FSLL-302	FSLL-302	FAT acid circulation low-low: sulfur shut-off (I-4)	SIF	0–1500	m³/h	1000	—	—	< 500 (10 s)	SIS	AI	—	002	SIS input from FT-302 (shared with the DCS indication and alarm)
AT-306	AIT-306	Stack SO_2 (dry)	AT	0–5000	ppm	292	—	500	—	DCS	AI	—	002	Analyzer range 0–5000 ppm; the DCS display span is narrower
ASHH-306	ASHH-306	Stack SO_2 high-high: sulfur shut-off (I-5)	SIF	0–5000	ppm	292	—	—	> 3000 (5 min)	SIS	AI	—	002	SIS input from AIT-306 (shared with the permit monitor and alarm)
AT-307	AIT-307	Stack SO_3 / acid mist	AT	0–100	mg/Nm³	10.5	—	40.0	—	DCS	AI	—	002	
XI-308	XI-308	Overall SO_2 conversion (calculated)	SOFT	98–100	%	99.80	—	—	—	DCS	—	—	002	DCS calculation
FI-309	FI-309	H_2SO_4 production, 100 % basis (calculated)	SOFT	0–160	t/h	125	—	—	—	DCS	—	—	—	DCS calculation; on the reference drawing only
K-101	K-101	Main air blower motor (≈4.2 MW)	MTR	—		—	—	—	—	DCS	DI + 2×DO	—	001	Run status, start, stop (assumed typical)
P-101A	P-101A	Sulfur pump A motor	MTR	—		—	—	—	—	DCS	DI + 2×DO	—	001	Assumed typical
P-101B	P-101B	Sulfur pump B motor	MTR	—		—	—	—	—	DCS	DI + 2×DO	—	001	Assumed typical
P-301A	P-301A	IPA acid pump A motor	MTR	—		—	—	—	—	DCS	DI + 2×DO	—	002	Assumed typical
P-301B	P-301B	IPA acid pump B motor (auto-start)	MTR	—		—	—	—	—	DCS	DI + 2×DO	—	002	Assumed typical
P-302A	P-302A	FAT acid pump A motor	MTR	—		—	—	—	—	DCS	DI + 2×DO	—	002	Assumed typical
P-302B	P-302B	FAT acid pump B motor (manual start)	MTR	—		—	—	—	—	DCS	DI + 2×DO	—	002	Assumed typical
H-101	H-101	Preheat burner package (start, stop, flame on)	MTR	—		—	—	—	—	DCS	DI + 2×DO	—	001	Vendor flame safeguard; interface assumed
HS-001	HS-001	ESD push-button (I-6)	HS	—		—	—	—	manual	SIS	DI	—	001	Tag from the EPC interlock table
UY-001A	HS-001	ESD output: stop K-101 (to MCC)	RLY	—		—	—	—	—	SIS	DO	—	—	Tag assumed; the model stops K-101 on ESD
UY-001B	HS-001	ESD output: stop H-101 (to burner package)	RLY	—		—	—	—	—	SIS	DO	—	—	Tag assumed; the model stops H-101 on ESD

3 · I/O count

Signals of part 2 added by system and type, plus a typical 20 % installed spare, rounded up to whole cards (channels per card assumed: AI 16, AO 8, DI 32, DO 32).

System	Signal	Used	Spare	Installed	Channels / card	Cards
DCS	AI	46	10	56	16	4
DCS	AO	10	2	12	8	2
DCS	DI	9	2	11	32	1
DCS	DO	16	4	20	32	1
SIS	AI	5	1	6	16	1
SIS	DI	3	1	4	32	1
SIS	DO	3	1	4	32	1
Total		92	21	113		11

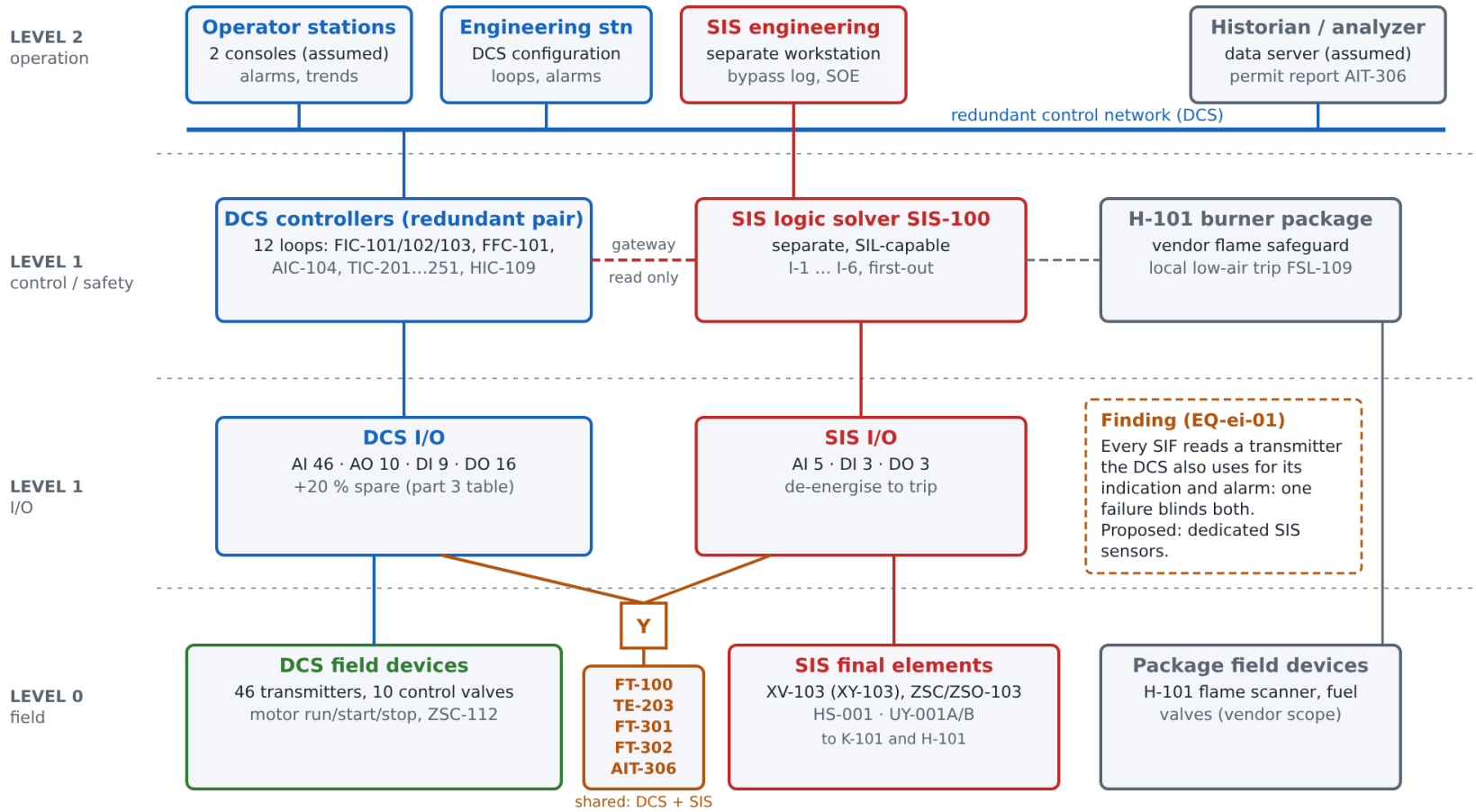
Worked calculation

DCS AI: 46 used; installed = ⌈46 × 1.20⌉ = ⌈55.2⌉ = 56; cards = ⌈56 / 16⌉ = 4. SIS AI: 5 used → 6 installed, 1 spare; the five dedicated SIS transmitters of EQ-ei-01 would make 10 used → 12 installed.

FSLL-100 in the FT-100 range

FT-100 0–300 kNm³/h, normal 238.5 kNm³/h (80 % of range). FSLL-100 at 90 kNm³/h = 30 % of range; an orifice sees $(0.30)^2$ = 9 % of its ΔP span there, at the edge of a typical 3–4 : 1 turndown (EQ-ei-05).

4 · DCS / SIS architecture



Assumed typical: SIS logic solver SIS-100 separate from the DCS, own I/O and engineering station, read-only gateway to the DCS. Amber: the five SIF sensors are the transmitters the DCS reads (finding EQ-ei-01).

5 · Control narrative

FIC-103 (AUTO, Kc 0.3, Ti 8 s) · Sets the production rate. CV: molten sulfur flow FT-103 (40.94 t/h); MV: FV-103 (fail closed). Normally AUTO with the set point entered by the operator; every SIS trip forces it to MAN at 0 % and closes XV-103. Its measured flow is the wild input of the ratio station FFC-101.

FFC-101 (AUTO) · Ratio station: multiplies the measured sulfur flow by 4.994 kNm³/t and writes the result as the remote set point of FIC-101, with a floor of 100 kNm³/h (K.airMin) so the furnace is never starved of air at low rate. No field I/O; the ratio is the operator's handle on excess O_2.

FIC-101 (CAS, Kc 0.25, Ti 5 s) · Combustion air to F-101. CV: FT-101 (204.5 kNm³/h at design); MV: FV-101 (fail open). Runs in CAS, slave of FFC-101: air follows sulfur. In AUTO the operator sets the air directly (start-up, sweep).

AIC-104 (AUTO, Kc 1.5, Ti 60 s) · SO_2 strength into bed 1. CV: analyzer AT-104 (12.00 % SO2); MV: the remote set point of FIC-102. Direct acting: SO_2 above set point asks for more dilution air. Master of a cascade; it tracks FIC-102 whenever the slave is not in CAS, and is forced to MAN by an SIS trip.

FIC-102 (CAS, Kc 0.25, Ti 5 s) · Dilution air. CV: FT-102 (34.1 kNm³/h at design); MV: FV-102 (fail open). Slave of AIC-104 in CAS; the fast flow loop absorbs blower and header disturbances before the slow analyzer sees them.

TIC-201 (AUTO, Kc 2, Ti 45 s) · Bed 1 inlet temperature. CV: TT-201 (420.0 °C); MV: TV-201, the hot-gas bypass round B-101 (fail closed: all gas through the boiler, the coolest feed). More bypass gives a hotter inlet.

TIC-211 (AUTO, Kc 3, Ti 60 s) · Bed 2 inlet. CV: TT-211 (445.0 °C); MV: TV-211, hot-side bypass of E-201 (fail open). Opening it heats bed 2 and leaves less heat for the return gas to bed 4.

TIC-221 (AUTO, Kc 3, Ti 60 s) · Bed 3 inlet. CV: TT-221 (440.0 °C); MV: TV-221, hot-side bypass of E-202 (fail open).

TIC-231 (AUTO, Kc 3, Ti 120 s) · Bed 4 inlet. CV: TT-231 on the return gas (425.0 °C); MV: TV-231, gas bypass of SH-201 (fail open), which changes how much heat reaches E-201. Bed 4 inlet has no exchanger of its own: the loop works through the heat beds 1–2 give up.

TIC-241 (AUTO, Kc 3, Ti 60 s) · Bed 5 inlet. CV: TT-241 (428.0 °C); MV: TV-241, gas bypass of E-204 (fail open).

TIC-251 (AUTO, Kc 1.5, Ti 40 s) · Gas to the interpass absorber T-301. CV: TT-251 (190.0 °C); MV: TV-251, gas bypass of E-203 (fail open: the gas stays above the acid dew point).

HIC-109 (MAN) · H-101 preheat burner firing, a hand controller (output only, 0–100 % of 30 MW), used for start-up heating; the burner package trips itself on flame failure or low air.

6 · Checks against the model and the other documents

Computed from the same data (node, tools/gen.js); the tab recomputes them live and adds one check that this PDF data equals the page.

Check	Instrument list	Model / other document	Result	Finding
Every DCS alarm (ALM) has an instrument in the list	21 of 21 alarm tags found	ALM in src/ui1.js: 21 tags	✓	
Every SIS trip is in the list with system = SIS	6 of 6 (I-1 ... I-6)	tripConditions(): 5 trips + ESD	✓	
Trip settings equal the model (K.trip*)	FSLL-100 90 · TSHH-203 660 · FSLL-301 800 · FSLL-302 500 · ASHH-306 3000	K.tripAir 90, K.tripT 660, 50 % of acid design, K.tripSO2 3000	✓	
Alarm and trip settings inside the instrument range	28 of 28 tags with a setting	Ranges: DCS spans / controller spans / analyzer range	✓	
Normal value inside the range and not beyond an alarm or trip	66 of 66 analog tags	Normal = model at design (snapDesign)	✓	
FSLL-100: SIS sensor independent of the BPCS	Sensor FT-100	Also used by FT-100 (alarm FI-100), FSL-109 (local function)	✗	Shared transmitter: see Safety layers part 5 and SRS part 5
TSHH-203: SIS sensor independent of the BPCS	Sensor TE/TT-203	Also used by TE/TT-203 (alarm TI-203)	✗	Shared transmitter: see Safety layers part 5 and SRS part 5
FSLL-301: SIS sensor independent of the BPCS	Sensor FT-301	Also used by FT-301 (alarm FI-301), FSL-301 (DCS function)	✗	Shared transmitter: see Safety layers part 5 and SRS part 5
FSLL-302: SIS sensor independent of the BPCS	Sensor FT-302	Also used by FT-302 (alarm FI-302)	✗	Shared transmitter: see Safety layers part 5 and SRS part 5
ASHH-306: SIS sensor independent of the BPCS	Sensor AT-306	Also used by AT-306 (alarm AIT-306)	✗	Shared transmitter: see Safety layers part 5 and SRS part 5

Check	Instrument list	Model / other document	Result	Finding
Valve fail positions match the P&ID sheets and the model	FV-101 FO · FV-102 FO · FV-103 FC · HV-109 FC · TV-201 FC · TV-231 FO · TV-211 FO · TV-221 FO · TV-241 FO · TV-251 FO	P&ID tab loop schedule (10 loops); instrument-air failure run	✓	
Every controller in makeCtrls() has its instruments in the list	12 of 12	makeCtrls()	✓	
Every DCS indicator and controller bubble (IND, BUB) is in the list	51 of 51	src/ui1.js IND + BUB	✓	
Instrument ranges equal the DCS display spans	47 of 48	IND[..].r, CRANGE, makeCtrls lo/hi	✗	AT-306 (DCS span 0–1500 ppm vs range 0–5000)
Temperature ranges cover the cold plant (ambient 30 °C)	7 of 25 TE/TT	K.Tamb; cold start-up heats from ambient	✗	18 ranges start at 100–300 °C: 201, 202, 203, 207, 211, 212, 213, 253, 221, 222, 223, 231, 232, 233, 241, 242, 243, 251
Instrument list I/O equals the I/O summary	92 points in the list	92 points in the summary	✓	

9 ✓ · 7 ✗ of 16. The five ✗ on SIF independence are the shared-transmitter finding (EQ-ei-01).

7 · Engineering queries

No.	Document	Subject	Finding	Proposed
EQ-ei-01	SO3-V1-IN-INL-001	SIF sensors shared with the DCS (FT-100, TE-203, FT-301, FT-302, AIT-306)	Each of the five SIS trips (FSLL-100, TSHH-203, FSLL-301, FSLL-302, ASHH-306) reads the same transmitter as the DCS indication and alarm; FT-301 also drives the P-301B auto-start and FT-100 the H-101 low-air interlock. One failed transmitter (frozen or reading on the safe side) defeats the alarm and the trip together. The instrument list carries them as SIS AI through a signal splitter (assumed).	Provide dedicated SIS transmitters (TE-203B, FT-301B, FT-100B, FT-302B; ASHH-306 treated as an environmental trip or given a second analyzer), wired to SIS-100, with the DCS reading a repeat. Spares in the SIS AI count cover one; the rest add SIS AI. Cross-reference: Safety layers part 5, SRS part 5, HAZOP recommendations.
EQ-ei-02	SO3-V1-IN-INL-001	Calibrated ranges not specified	No instrument datasheet exists for this unit. The list takes each range from the DCS display span (src/ui1.js IND / CRANGE), the controller span in makeCtrls(), or the analyzer range stated in measure().	Confirm every calibrated range on the instrument datasheets (Rev A) and re-issue the list; until then the ranges are marked as DCS spans, i.e. assumed.
EQ-ei-03	SO3-V1-IN-INL-001	AIT-306 DCS span narrower than the analyzer range	The stack SO_2 analyzer is ranged 0–5000 ppm (measure()), but the DCS display span of AIT-306 is 0–1500 ppm. The ASHH-306 setting (3000 ppm) is off the top of the DCS bar and trend.	Set the DCS span of AIT-306 to the analyzer range 0–5000 ppm (keep a 0–1500 ppm trend pen for the permit), so the operator sees the approach to the trip.
EQ-ei-04	SO3-V1-IN-INL-001	Converter temperature ranges start at 100–300 °C	18 of 25 temperature transmitters (bed inlet, middle and exit TCs, TI-207, TI-253, TI-251) have spans starting at 100–300 °C, but a cold start-up heats the converter from ambient (K.Tamb 30 °C) and the start permissives need TI-201/202 ≥ 400 °C and TI-232 ≥ 360 °C to be seen rising.	Range the transmitters 0–700 °C (beds) and 0–400 °C (TT-251); keep the narrow controller spans in the DCS for tuning only.
EQ-ei-05	SO3-V1-IN-INL-001	FSLL-100 trip low in the FT-100 orifice range	FSLL-100 trips at 90 kNm³/h, 30 % of the 0–300 kNm³/h range of FT-100: an orifice gives only 9 % of its differential-pressure span there ($\Delta P \propto Q^2$), where accuracy is poor (typical orifice turndown 3–4 : 1).	Dedicated SIS flow transmitter (see EQ-ei-01) ranged for the trip (e.g. 0–150 kNm³/h), or a low-range DP cell; confirm with the flow element calculation.
EQ-ei-06	SO3-PID-001 rev A	Position switches for XV-103 and HV-112 not on the P&ID	The start permissive 'HV-112 closed' and the proof of XV-103 closure need position switches; the model reads the valve states directly and neither sheet shows ZSC-112 or ZSC/ZSO-103. The list carries them as assumed (1 DCS DI, 2 SIS DI).	Add ZSC-112 (on the LC valve) and ZSC-103 / ZSO-103 to SO3-PID-001 at rev B and wire ZSC-103 to SIS-100.
EQ-ei-07	SO3-PID-001/002 rev A	Eight DCS instruments not on the EPC sheets	FI-109, FI-253 and FI-309 are on the reference drawing only; PI-200, TI-207, TI-253, TI-254 and TI-255 are on the DCS but on neither drawing (P&ID tab part 5). They are in this list with P&ID sheet '—'.	Add them to SO3-PID-001/002 at rev B (already an open item of the P&ID tab part 6).
EQ-ei-08	SO3-V1-IN-INL-001	Second sulfur shut-off goes through the DCS	Every trip closes XV-103 (SIS) and forces FIC-103 to MAN 0 % so FV-103 also closes, but that second action is a DCS command over the gateway, not an SIS output.	Add an SIS solenoid XY-103B on FV-103 (Safety layers part 5): +1 SIS DO, inside the 20 % spare.
EQ-ei-09	SO3-V1-IN-INL-001	Start permissives read DCS measurements	The sulfur admission permissives (TI-106, TI-201, TI-202, TI-232, FI-301, FI-302, HV-112, FIC-103 MAN ≤ 25 %) are checked by the SIS in the model but come from DCS transmitters and a DCS controller state.	Define whether permissives are SIS logic (hardwired or SIS transmitters) or DCS logic releasing an SIS reset; document in the SRS and the control narrative.

8 · Assumptions

- Calibrated ranges = DCS display spans (src/ui1.js), controller spans (makeCtrls) or analyzer range (AIT-306 0–5000 ppm); to be confirmed on datasheets.
- Motors and the H-101 package: 1 DI running + 2 DO start/stop each; XV-103 with ZSC/ZSO-103 to the SIS; ZSC-112 on HV-112 to the DCS; ESD outputs UY-001A/B; FSL-109, HV-109, XY-103, UY-001A/B tags assumed.
- SIS inputs of the five trips taken from the shared transmitters through a splitter (as modelled); valve position feedback via positioner HART, no separate I/O.
- 20 % installed spare and channels per card AI 16, AO 8, DI 32, DO 32 (typical); two operator consoles, a historian and a 1oo2D-class logic solver are typical, not modelled.